



Specialized Topics of Theoretical Physics

Majorana Modes in the Machine – Simulating Topological Phases with Quantum Circuits and AI

Finite-Size Physics & Majorana Edge Modes

Zhenming Shi
[3776685]

Supervisor: Prof. Dr. Bernd Rosenow

Contents

1	Introduction	1
2	Theory: edge modes of the finite open chain	1
2.1	BdG form of the finite chain	1
2.2	Edge mode and localisation length at the sweet spot	1
2.3	Hybridisation splitting of the zero mode	2
2.4	Exact zero-energy lines for $t \neq \Delta$	2
2.5	The finite-size quantization rule	3
3	Methods and results	3
3.1	OBC spectrum and the two mid-gap levels	3
3.2	Exponential protection: E_0 versus L	4
3.3	Away from the sweet spot: Majorana lines and parity switches	4
3.4	The sign-filter artefact	5
4	On the AI collaboration	5

1 Introduction

The 1D Kitaev chain is a toy model of a topological superconducting wire, showing how Majorana zero modes arise at its ends. **Part 1** studied the infinite chain with periodic boundaries, where the phase $|\mu| < 2t$ is fixed by a bulk invariant. A real wire is finite and open, and that finiteness is what turns ideal zero modes into something measurable: this report treats the spectrum of the finite open chain, the edge modes and their exponential localisation, the hybridisation splitting $E_0(L)$, and the sign-filter artefact that appears once the splitting drops below machine precision. Beyond the usual sweet-spot ($t = \Delta$) treatment, the analytic backbone is the exact finite- L theory of Leumer *et al.* [4, 5]: a determinant criterion locating the L discrete “Majorana lines” on which the splitting vanishes *exactly*, and a quantization rule treating bulk and edge states on the same footing.

2 Theory: edge modes of the finite open chain

2.1 BdG form of the finite chain

We consider the Kitaev chain on L sites with open boundary conditions,

$$H_K = -\mu \sum_{j=1}^L c_j^\dagger c_j - t \sum_{j=1}^{L-1} (c_j^\dagger c_{j+1} + \text{h.c.}) + \Delta \sum_{j=1}^{L-1} (c_j^\dagger c_{j+1}^\dagger + \text{h.c.}), \quad (2.1)$$

with real t, Δ and the symmetric (“sweet”) point $t = \Delta$. In the Nambu basis $\Psi = (c_1, \dots, c_L, c_1^\dagger, \dots, c_L^\dagger)^T$ it becomes a $2L \times 2L$ quadratic form,

$$H_K = \frac{1}{2} \Psi^\dagger M \Psi, \quad M = \begin{pmatrix} h & d \\ -d & -h \end{pmatrix}, \quad h_{ij} = -\mu \delta_{ij} - t(\delta_{i,j+1} + \delta_{i+1,j}), \quad d_{ij} = \Delta(\delta_{j,i+1} - \delta_{i,j+1}), \quad (2.2)$$

with h real symmetric tridiagonal and d real antisymmetric, carrying $+\Delta$ on its upper diagonal so that $\frac{1}{2} \Psi^\dagger M \Psi$ reproduces the pairing term of (2.1). The particle–hole operator $\mathcal{C} = \tau_x \mathcal{K}$ (τ_x acting on the particle–hole blocks, $\mathcal{K}$ complex conjugation) gives $\mathcal{C} M \mathcal{C}^{-1} = -M$, so $M\psi = E\psi$ implies $M(\mathcal{C}\psi) = -E(\mathcal{C}\psi)$: the $2L$ eigenvalues come in $\pm E$ pairs, each pair *one* physical quasiparticle rather than two, so each pair must be counted once.

2.2 Edge mode and localisation length at the sweet spot

The Ansatz $\psi_j \sim \lambda^j$ reduces the bulk BdG equations to a 2×2 block with $A = -\mu - t(\lambda + \lambda^{-1})$ on the diagonal and $B = -\Delta(\lambda - \lambda^{-1})$ off it; $B^2 - A^2 = 0$ is the characteristic equation

$$[-\mu - t(\lambda + \lambda^{-1})]^2 - \Delta^2(\lambda - \lambda^{-1})^2 = 0. \quad (2.3)$$

At $t = \Delta$ this factorises as $(A - B)(A + B) = 0$, giving the branches $\lambda = -\mu/2t$ and $\lambda = -2t/\mu$; the left edge mode is the decaying one, $|\lambda| < 1$:

$$\psi_j^L \propto \lambda^{j-1}, \quad \lambda = -\frac{\mu}{2t}, \quad \xi = \frac{1}{\ln(2t/|\mu|)}, \quad (2.4)$$

which exists precisely inside the topological phase $|\mu| < 2t$; ξ is the coherence length and $\psi_j^R \propto \lambda^{L-j}$ the mirrored right edge mode. The decay rate is $|\lambda| = |\mu|/2t$; the only structure on the exponential is the sign of λ , a $(-1)^j$ staggering present for $\mu > 0$ and absent for the $\mu < 0$ used throughout. For $t \neq \Delta$ the roots of (2.3) can be genuinely complex and the splitting oscillates around the envelope [1, 3].

2.3 Hybridisation splitting of the zero mode

In the infinite chain ψ^L and ψ^R are degenerate at exactly $E = 0$; in a finite chain they overlap and the degeneracy is lifted by $\delta E \approx |\langle \psi^L | H | \psi^R \rangle| / (\|\psi^L\| \|\psi^R\|)$. The matrix element is boundary-dominated: both modes annihilate the infinite-chain Hamiltonian, and the open chain differs from it only by the bonds removed at the ends, so $H|\psi^R\rangle$ is non-zero only where the right mode has leaked to the left edge, of size λ^{L-1} . The normalisation $\|\psi^L\|^2 \rightarrow 1/(1 - \lambda^2)$ only sets the prefactor, fixed by the exact finite-chain calculation at $t = \Delta$:

$$E_0(L) \approx 2t (1 - \lambda^2) \left(\frac{|\mu|}{2t} \right)^L, \quad |\lambda| = \frac{|\mu|}{2t}. \quad (2.5)$$

Three predictions follow: $E_0 = 0$ exactly at $\mu = 0$ for any L ; exponential decay in the interior with semilog slope $\ln |\lambda| = -1/\xi$; and, as $|\mu| \rightarrow 2t$, $|\lambda| \rightarrow 1$ and $\xi \rightarrow \infty$, so (2.5) — which assumes $L \gg \xi$ — breaks down and the splitting stops being exponentially small, saturating instead at the finite-size gap $\simeq \pi t/L$. The zero mode is therefore never protected near the transition. The true eigenstates are $(\psi^L \pm \psi^R)/\sqrt{2}$ at $\pm E_0$: the two edge Majoranas jointly form a single non-local fermion, and $E_0(L)$ measures how far a finite system is from hosting true zero modes.

2.4 Exact zero-energy lines for $t \neq \Delta$

Equation (2.5) is asymptotic ($L \gg \xi$) and tied to the sweet spot. The sharper question — *when is E_0 exactly zero at finite L ?* — has a closed answer for all t, Δ, μ , due to Leumer *et al.* [4, 5]. Split each fermion into Majorana operators $c_j = (\gamma_j^A + i\gamma_j^B)/2$ and order them by species, $\psi_c = (\gamma_1^A, \dots, \gamma_L^A, \gamma_1^B, \dots, \gamma_L^B)^T$. Since (2.1) contains no $\gamma^A \gamma^A$ or $\gamma^B \gamma^B$ terms, the BdG matrix in this *chiral basis* is block off-diagonal, $H_K = \frac{1}{4} \psi_c^\dagger H_c \psi_c$ with

$$H_c = \begin{pmatrix} 0 & h_c \\ h_c^\dagger & 0 \end{pmatrix}, \quad (h_c)_{ij} = -i\mu \delta_{ij} - b \delta_{j,i+1} + a \delta_{i,j+1}, \quad a = i(\Delta - t), \quad b = i(\Delta + t), \quad (2.6)$$

tridiagonal but *not* Hermitian. (Ref. [5] uses the opposite pairing sign, which transposes h_c ; determinants, and hence everything below, are unchanged.) Zero energy is then detected by a determinant rather than the full spectrum: the block structure gives $\det H_c = \det(-h_c h_c^\dagger) = (-1)^L |\det h_c|^2$, so $E = 0$ occurs *iff* $\det h_c = 0$. The leading principal minors of a tridiagonal matrix obey $D_j = -i\mu D_{j-1} + ab D_{j-2}$ with $ab = t^2 - \Delta^2$, and the Ansatz $D_j \propto (ir)^j$ turns this into $r^2 + \mu r + (t^2 - \Delta^2) = 0$, hence

$$\det h_c = i^L \frac{r_+^{L+1} - r_-^{L+1}}{r_+ - r_-}, \quad r_\pm = \frac{1}{2} \left(-\mu \pm \sqrt{\mu^2 + 4(\Delta^2 - t^2)} \right), \quad (2.7)$$

which vanishes *iff* $r_+^{L+1} = r_-^{L+1}$ (for the degenerate case $r_+ = r_-$ the limit is $\det h_c = i^L (L+1) r_+^L$, zero only at $r_+ = 0$). Using $r_+ r_- = t^2 - \Delta^2$ and parametrising $\mu = 2\sqrt{t^2 - \Delta^2} \cos \theta$ gives $r_\pm = -\sqrt{t^2 - \Delta^2} e^{\mp i\theta}$, so the condition reduces to $\sin[(L+1)\theta] = 0$ with $\sin \theta \neq 0$, i.e. $\theta_n = n\pi/(L+1)$:

$$\mu_n = 2\sqrt{t^2 - \Delta^2} \cos\left(\frac{n\pi}{L+1}\right), \quad n = 1, \dots, L, \quad t^2 \geq \Delta^2. \quad (2.8)$$

A finite open chain therefore carries *exact* zero modes only on L discrete “Majorana lines” [4], and only for $|t| \geq |\Delta|$ — the sole exception being $\mu = 0$ for odd L , where $n = (L+1)/2$ kills the cosine for any t, Δ . Elsewhere in the topological phase E_0 is small but strictly finite. Two consequences matter. At the sweet spot the square root collapses and all L lines merge into $\mu = 0$: there $\{r_+, r_-\} = \{0, -\mu\}$ and $|\det H_c| = \mu^{2L}$, vanishing only at $\mu = 0$ — hence a smooth $E_0(\mu)$ with a single exact zero. For

$|\Delta| < |t|$, instead, E_0 oscillates beneath its envelope and touches zero L times, though only inside the sub-interval $|\mu| \leq 2\sqrt{t^2 - \Delta^2}$ of the topological window; each crossing is a fermion-parity switch of the ground state [1, 3].

The mode on a line is exact too: $h_c^\dagger \vec{v}_A = 0$ reads $b\xi_{j-1} + i\mu\xi_j - a\xi_{j+1} = 0$, a Fibonacci recursion whose solution with $\xi_0 = 0$ and $\mu = \mu_n$ is [5]

$$\xi_j \propto (-1)^{j-1} \frac{\sin(j\theta_n)}{\sin\theta_n} e^{-(j-1)/\xi_{\text{loc}}}, \quad \xi_{\text{loc}} = 2/\ln\left|\frac{t+\Delta}{t-\Delta}\right|, \quad (2.9)$$

a pure- γ^A Majorana bound to one edge, its γ^B partner ($a \leftrightarrow b$, $\mu \rightarrow -\mu$) decaying from the other. The envelope is exponential, but the chemical potential survives inside $\sin(j\theta_n)$ as a real-space *oscillation* of the wavefunction — the lattice analogue of the factor e^{ik_0j} absent in (2.4).

2.5 The finite-size quantization rule

The determinant settles only the $E = 0$ question. The full finite- L spectrum follows from squaring the problem to $h_c h_c^\dagger \vec{v} = E^2 \vec{v}$, whose amplitudes need a four-wave Ansatz with *two* momenta $k_{1,2}$ at the same energy; both lie on the bulk dispersion and the open boundaries quantise them [4, 5]:

$$\cos k_1 + \cos k_2 = -\frac{\mu t}{t^2 - \Delta^2}, \quad \frac{\sin^2[(L+1)k_\Sigma]}{\sin^2[(L+1)k_\Delta]} = \frac{1 + (\Delta/t)^2 \cot^2 k_\Delta}{1 + (\Delta/t)^2 \cot^2 k_\Sigma}, \quad (2.10)$$

with $k_{\Sigma,\Delta} = (k_1 \pm k_2)/2$. Real $k_{1,2}$ are bulk standing waves, complex $k_{1,2}$ the sub-gap edge states, and three finite-size lessons follow. (i) Since superconductivity puts $\pm\mu$ into the particle and hole blocks of (2.2), μ cannot act as a rigid energy shift: it enters the quantization itself, $k_n = k_n(\mu, t, \Delta)$, so the spacing is generically non-equidistant and every level drifts as μ is swept; the particle-in-a-box values $n\pi/(L+1)$ survive only at the level crossings and on the zero-energy lines (2.8) [5]. (ii) Edge modes are the bulk solutions that go missing: some real- k solutions vanish inside the window and re-enter with complex momentum. Eq. (2.4) is one, $k = i/\xi$ at $\mu < 0$ — purely evanescent, hence the monotonic sweet-spot splitting, whereas a generic $\text{Re } k$ at $t \neq \Delta$ makes the two evanescent waves interfere. (iii) The boundary conditions exclude $k = 0, \pi$, so the gap closing at $|\mu| = 2t$ is unreachable at finite L : the spectrum only pinches to a finite-size gap.

3 Methods and results

All results are exact diagonalisations of the BdG matrix (2.2) at $t = \Delta = 1$ unless stated otherwise, obtained with the symmetric eigensolver `numpy.linalg.eigvalsh` in double precision. Quasiparticle energies are taken as $\text{sort}(|E|)$ keeping one entry per $\pm E$ pair, rather than by filtering on the sign of E .

3.1 OBC spectrum and the two mid-gap levels

Figure 1 sweeps μ through both critical points at $L = 30$. A near-zero branch spans the whole topological window; beyond $|\mu| = 2t$ no sub-gap state remains. At $\mu = -t$ it sits at $E_0 = 1.397 \times 10^{-9}$, matching $2t(1 - \frac{1}{4})(\frac{1}{2})^{30}$ from Eq. (2.5) to six significant figures; at $\mu = 0$ it is 2.5×10^{-16} , zero to double precision. The branch *is* the hybridised Majorana pair; the rounded merger at $|\mu| = 2t$ is the finite-size shadow of the bulk transition (see [Schewtschik, Part 1]), exact closing being forbidden at finite L by lesson (iii) of Sec. 2.5. Figure 2 resolves this state by state: the $L = 20$ spectrum is gapped at $\mu = -3t$ but carries two near-zero eigenvalues inside the gap at $\mu = -t$ — not two excitations but the $\pm E_0$ pair of a *single* level, the non-local fermion built from the two edge Majoranas.

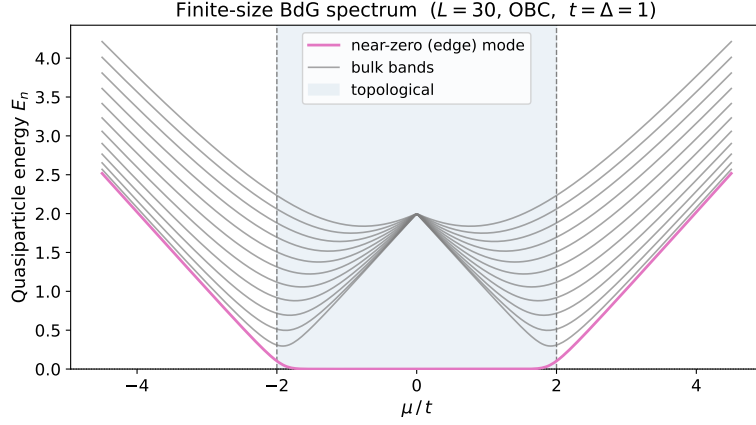


Figure 1: Lowest twelve quasiparticle levels of the $L = 30$ open chain vs μ : the near-zero edge branch spans the topological window and merges into the bulk continuum at $|\mu| = 2t$.

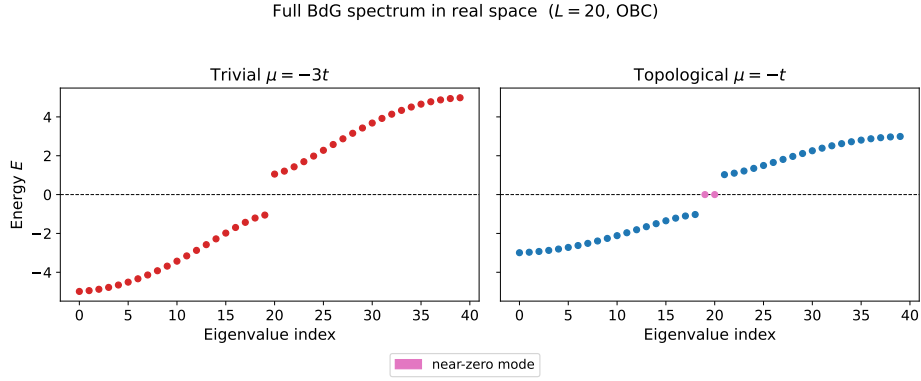


Figure 2: Full BdG spectrum of the $L = 20$ open chain: trivial phase (left, no sub-gap states) versus topological phase (right, a single $\pm E_0$ near-zero pair isolated inside the gap).

3.2 Exponential protection: E_0 versus L

Figure 3 shows $E_0(L)$ for $\mu/t = -0.5, -1, -1.5$ against Eq. (2.5) (dashed). The data are straight on a semilog scale with fitted slopes reproducing $\ln |\lambda|$, and the ratios $E_0/|\lambda|^L$ reproduce the prefactor $2t(1-\lambda^2)$, ruling out the bare estimate $2t|\lambda|^L$ (Table 1). The slope measures the coherence length, $\xi = 1/|\text{slope}|$: 0.7 sites at $\mu = -0.5t$ but 3.4 at $\mu = -1.5t$, so protection weakens toward the phase boundary. Where the data leave the theory lines the splitting has fallen below machine precision (at $\mu = -0.5t$, $L = 40$ it should be 1.55×10^{-24}) and the measured values saturate in the shaded band.

3.3 Away from the sweet spot: Majorana lines and parity switches

Figure 4 tests Eq. (2.8) on one $L = 20$ chain at two pairings. At $\Delta = t$ the splitting is a smooth valley whose only exact zero is $\mu = 0$, where all Majorana lines coalesce. At $\Delta = 0.4t$ it instead oscillates beneath

Table 1: Fits to Fig. 3: slopes of $\ln E_0$ vs L over the even- L grid 6, 8, ..., 24; ratios at $L = 20, 40, 100$ for $\mu/t = -0.5, -1, -1.5$ respectively (beyond those L the splitting has sunk below machine precision and the ratio is meaningless).

μ/t	fitted slope	$\ln \lambda $	$E_0/ \lambda ^L$	$2t(1-\lambda^2)$	$\xi = 1/ \text{slope} $
-0.5	-1.3866	-1.3863	1.8748	1.875	0.72
-1.0	-0.6932	-0.6931	1.5000	1.500	1.44
-1.5	-0.2907	-0.2877	0.8735	0.875	3.44

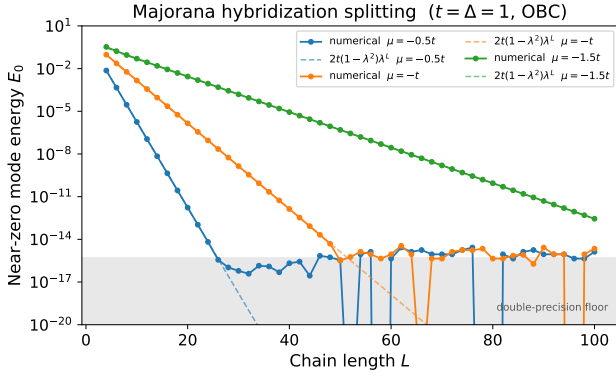


Figure 3: E_0 vs L with Eq. (2.5) dashed: slope $-1/\xi$, saturating in the shaded precision band.

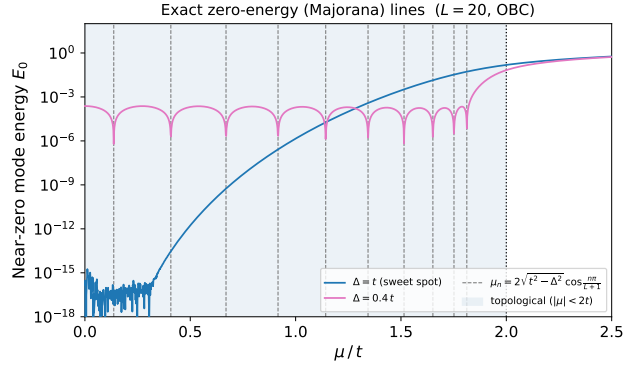


Figure 4: $E_0(\mu)$ at $L = 20$: smooth at $\Delta = t$; oscillating at $\Delta = 0.4t$ with exact zeros on the lines μ_n of Eq. (2.8) (dashed).

its envelope and dips to an exact zero at each of the ten predicted $\mu_n > 0$ (dashed), with no fit parameter. Diagonalisation at $\mu = \mu_n$ gives E_0 between 3×10^{-18} and 3×10^{-16} , zero to double precision, against $E_0 \approx 2 \times 10^{-4}$ midway between lines — twelve orders of magnitude apart. Each μ_n is a level crossing at which the ground-state fermion parity switches. Since a real wire is never fine-tuned to $t = \Delta$, this oscillatory splitting — the lattice cousin of the Majorana oscillations of semiconductor nanowires [1] — is the generic finite-size fingerprint, and Fig. 3 its cleanest special case.

3.4 The sign-filter artefact

Extracting the “positive spectrum” naively as `evals[evals >= 0]` produces dense spikes over most of the topological window at large L (Fig. 5, left), while $|E|$ on the *same* eigenvalues is smooth (right). The spikes are a floating-point artefact, not physics [2]: the near-zero eigenvalue drops below machine precision once

$$E_0 < \varepsilon_{\text{mach}} \iff |\mu| \lesssim 2t \varepsilon_{\text{mach}}^{1/L}, \quad (3.1)$$

i.e. $|\mu|_{\text{thresh}} \approx 0.62$ at $L = 30$ but ≈ 1.41 at $L = 100$ for $\varepsilon_{\text{mach}} = 5 \times 10^{-16}$. Inside that window rounding can store the near-zero eigenvalue as a tiny *negative* number $\sim -10^{-16}$; the filter drops it and reports the lowest *bulk* level as “ E_0 ” — at $\mu = -1.38t$, $L = 100$ it returns 0.622 where $|E|$ gives 3.1×10^{-15} . Since the window *widens* with L , the spikes worsen for longer chains even though Eq. (2.5) says longer chains protect the mode better — the counter-intuitive observation that motivated this analysis. The lesson is to extract a symmetry-pinned quantity so that it survives ε -sized violations of that symmetry — here $|E|$ together with pair deduplication, which is the extraction used throughout this report.

4 On the AI collaboration

An LLM mined the block notes and reproduced the chiral-basis determinant argument of Sec. 2.4 from Ref. [5] in the convention of Eq. (2.1). Three corrections came out of that exchange: the measured ratios (Table 1) forced the $(1 - \lambda^2)$ prefactor onto an initially bare $2t|\lambda|^L$; a physical reading of the large- L spikes was dropped once Eq. (2.5) was seen to predict the opposite trend; and an adversarial re-check caught a sign slip in Eq. (2.3) that had propagated into λ .

Compressing the material to the page limit proved the more instructive half. What to cut is a physics judgement as much as an editorial one — a step earns its place only if the argument cannot be reconstructed without it — and settling that forced an explicit view of what each section is for. Presentation is a technique

Sign-filter artefact vs stable $|E|$ extraction for E_0 ($L = 100$, $t = \Delta = 1$, OBC)

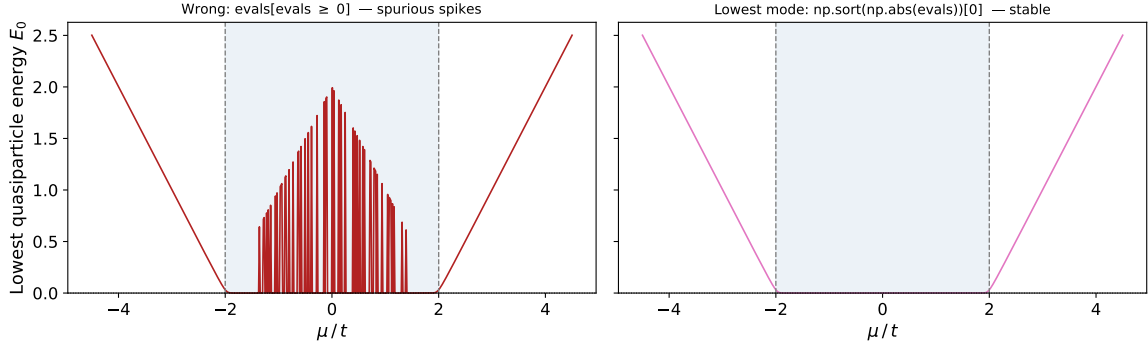


Figure 5: Same eigenvalues, two extractions ($L = 100$): the sign filter (left) fabricates bulk-scale spikes over the window of Eq. (3.1); $|E|$ (right) resolves the near-zero mode.

in its own right, to be practised like the physics: redrafting against the model sharpened the register, showing where a hedge is honest and where it is padding, and how a scientific sentence is built to carry one claim at a time. Every number was re-verified against the code; the draft is to be revised by the author.

References

- [1] S. Das Sarma, Jay D. Sau, and Tudor D. Stanescu. Splitting of the zero-bias conductance peak as smoking gun evidence for the existence of the Majorana mode in a superconductor–semiconductor nanowire. *Physical Review B*, 86:220506, 2012. doi: 10.1103/PhysRevB.86.220506.
- [2] David Goldberg. What every computer scientist should know about floating-point arithmetic. *ACM Computing Surveys*, 23(1):5–48, 1991. doi: 10.1145/103162.103163.
- [3] Suraj S. Hegde and Smitha Vishveshwara. Majorana wave-function oscillations, fermion parity switches, and disorder in Kitaev chains. *Physical Review B*, 94:115166, 2016. doi: 10.1103/PhysRevB.94.115166.
- [4] Nico Leumer, Magdalena Marganska, Bhaskaran Muralidharan, and Milena Grifoni. Exact eigenvectors and eigenvalues of the finite Kitaev chain and its topological properties. *Journal of Physics: Condensed Matter*, 32(44):445502, 2020. doi: 10.1088/1361-648X/ab8bf9.
- [5] Nico Gerhard Leumer. *Spectral and transport signatures of 1d topological superconductors of finite size in the sub- and supra-gap regime: An analytical study*. PhD thesis, Universität Regensburg, 2021. <https://epub.uni-regensburg.de/50905/>.